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An Airport Operation Assurance Risk Identification Model Based on the DBSCAN Algorithm

Mingjie Guan
Airport Research Institute
China Academy of Civil Aviation Science and Technology
Beijing, China
hangke126@126.com

Zhengning Yu
Airport Research Institute
China Academy of Civil Aviation Science and Technology
Beijing, China
yuzhn@mail.castc.org.cn

***Abstract*—With the rapid growth of the aviation industry, airport operation assurance risk management has become a critical aspect of ensuring aviation safety. The Voluntary Reporting System (VRS) serves as a vital component of risk management, providing a wealth of textual data for risk identification. This paper proposes a text analysis approach based on the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm, capable of automatically identifying sources of hazards from voluntary reports, assessing the likelihood and consequence of risks, and further determining risk levels. An automated risk assessment model was constructed to improve the efficiency and safety of airport operation assurance. Additionally, the model's output data can be integrated into the risk management processes that drive the digital twin airport as a mapping of physical entities in the information world.**

Keywords—Airport operation assurance, Risk identification, DBSCAN algorithm, Text analysis, Voluntary Reporting System (VRS), Digital Twin Airport

I. Introduction

Airports, as critical nodes in the global aviation transportation network, play a pivotal role in ensuring the safety and efficiency of operations, which is crucial for the stable development of the entire aviation industry. With the continuous growth of air traffic, the types and complexity of risks faced by airport operation assurance are increasing, including but not limited to equipment failures, adverse weather conditions, security threats, and operational errors. The existence of these risks not only threatens the safety of passengers and personnel but may also result in flight delays, economic losses, and even more severe consequences. Therefore, effective risk management [1] has become an indispensable part of airport operations.

In the risk management process, risk identification is the primary and critical step. Traditional risk identification methods rely on expert experience and qualitative assessments, which are not only time-consuming but may also be subject to subjective judgments, making it difficult to comprehensively cover all potential risks. With the development of information technology, an increasing number of airports are exploring the use of data analysis techniques to improve the efficiency and accuracy of risk identification.

The Voluntary Reporting System (VRS) [2] serves as an important source of risk information, collecting a large number of reports on potential issues and abnormal situations in airport operations. These reports are often in textual form, containing rich unstructured data. Extracting useful information from these text data and transforming it into a powerful tool for risk management is a pressing issue.

The DBSCAN algorithm [3], as a density-based clustering [4] algorithm, can effectively handle complex datasets with noise and identify the natural structure of the data. It does not require a predetermined number of clusters, automatically identifies the number of clusters based on the characteristics of the data, and can identify clusters of any shape. These characteristics make the DBSCAN algorithm a powerful tool for identifying risks from voluntary reporting text data.

This paper aims to study and construct an airport operation assurance risk identification model based on the DBSCAN algorithm. The model can automatically extract key information from voluntary reports and identify potential sources of risks. Such data can serve as a mapping of physical entities in the information world, applied in digital twin airport [5] models as a driving data source for risk management. Through this approach, the efficiency and accuracy of airport operation assurance risk management can be enhanced, providing scientific decision support for airport operations to ensure the safety and smoothness of aviation transportation.

II. Literature Review

A. Existing Approaches to Airport Operation Assurance Risk Management

Airport operation assurance risk management involves the systematic identification, assessment, and mitigation of potential hazards and vulnerabilities in airport operations to ensure the safety, security, and efficiency of air travel. Over the years, various approaches and methodologies have been developed to address the complex challenges associated with managing risks in airport environments. This section provides an overview of some of the existing approaches to airport operation assurance risk management:

1) Traditional Risk Management Frameworks

Traditional risk management frameworks, such as the International Organization for Standardization (ISO) 31000 [6] and the Project Management Institute's (PMI) Project Risk Management Framework, provide a structured approach to identifying, assessing, and managing risks. These frameworks typically involve processes such as risk identification, risk analysis, risk evaluation, risk treatment, and risk monitoring and control. While these frameworks

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offer a systematic methodology for risk management, they may lack specificity in addressing the unique challenges of airport operations.

2) Safety Management Systems (SMS)

Safety Management Systems (SMS) [7] are proactive risk management systems implemented by aviation organizations to identify and mitigate safety risks. SMS typically involve the establishment of safety policies, risk assessment processes, safety reporting systems, safety assurance activities, and safety promotion initiatives. While SMS primarily focus on safety risks, they can also be adapted to address broader operational risks in airport environments. A typical SMS process can be summarized as shown in Fig. 1.

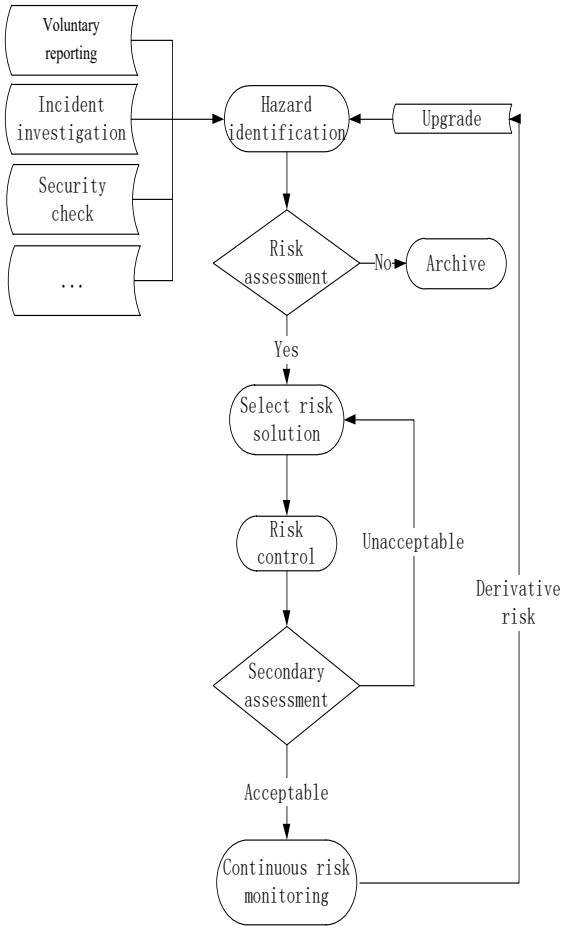


Fig. 1. Flowchart of Safety Management Systems (SMS).

3) Human Factors Engineering

Human factors engineering focuses on understanding the interaction between humans and systems to optimize performance, safety, and usability. In the context of airport operation assurance, human factors engineering principles are applied to design airport facilities, equipment, procedures, and interfaces to minimize human error and enhance operational safety. This approach recognizes the critical role of human factors in aviation safety and seeks to mitigate risks associated with human error.

4) Data-Driven Risk Management

With the advent of big data analytics and machine learning technologies, airports are increasingly leveraging data-driven approaches to risk management. By analyzing large volumes of operational data, including flight data, weather data, maintenance records, and safety reports, airports can identify patterns, trends, and anomalies indicative of potential risks. Data-driven risk management enables airports to make more informed decisions and allocate resources more effectively to mitigate risks.

5) Collaborative Risk Management

Collaborative risk management involves fostering collaboration and information sharing among various stakeholders in the aviation industry, including airlines, airport operators, air traffic control agencies, regulatory authorities, and industry associations. By collaborating on risk identification, assessment, and mitigation efforts, stakeholders can leverage their collective expertise and resources to address common challenges and enhance overall operational resilience.

6) Emerging Technologies

Emerging technologies such as artificial intelligence (AI) [8], Internet of Things (IoT) [9], and blockchain are increasingly being explored for their potential to enhance airport operation assurance risk management. AI-powered predictive analytics can forecast potential risks and recommend proactive mitigation measures, while IoT sensors can provide real-time monitoring of airport infrastructure and operations. Blockchain technology can facilitate secure and transparent data sharing among stakeholders, enhancing trust and collaboration in risk management efforts.

B. The Voluntary Reporting System (VRS)

The Voluntary Reporting System (VRS) is pivotal in airport operation assurance for identifying potential hazards. It encourages frontline personnel to report safety concerns voluntarily, fostering transparency and accountability. VRS reports offer qualitative and quantitative data on various operational aspects, aiding in a nuanced understanding of risks. A typical form of VRS is shown in Fig. 2.

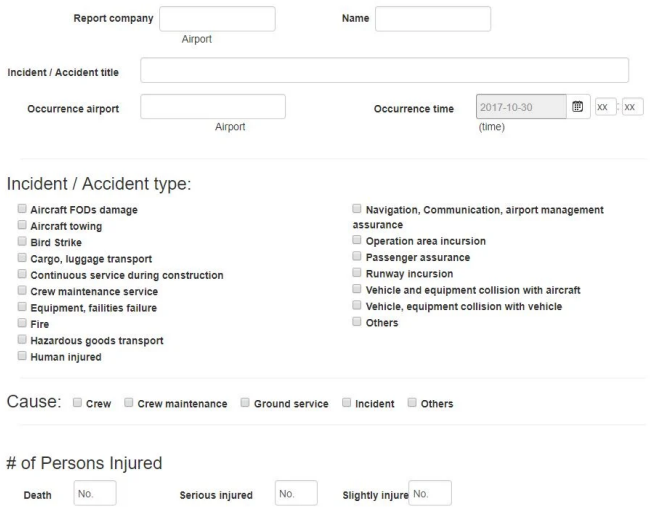


Fig. 2. A typical form of VRS.

Timely reporting through VRS serves as an early warning system, enabling proactive measures to mitigate

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risks. The system supports continuous improvement by identifying recurring issues and root causes, fostering organizational learning. Complementing regulatory oversight, VRS provides bottom-up insights into safety risks.

By empowering frontline personnel to report without fear, VRS cultivates a safety culture of open communication and collaboration. This encourages proactive risk identification, enhancing the overall safety performance of airport operations.

C. Utilization of text analysis techniques in risk identification

In the domain of airport operation assurance, the integration of text analysis techniques has emerged as a powerful tool for identifying and assessing potential risks embedded within unstructured textual data. This section explores the multifaceted utilization of text analysis techniques in risk identification processes:

Text preprocessing serves as the initial step, involving techniques like tokenization, stemming, and stop-word removal to standardize and refine raw textual data. By ensuring consistency and relevance, this process lays a solid foundation for subsequent analysis.

Sentiment analysis offers insights into the sentiments expressed within textual data, aiding in discerning positive, negative, or neutral tones associated with reported incidents or observations. Such analysis helps in pinpointing areas of concern or potential safety hazards.

Topic modeling techniques, such as Latent Dirichlet Allocation (LDA) [10] or Non-Negative Matrix Factorization (NMF) [11], unravel latent themes or topics prevalent within large datasets of textual information. By identifying recurring themes or patterns, topic modeling facilitates the detection of potential risks or safety issues.

Named Entity Recognition (NER) [12] techniques enable the identification and classification of named entities within textual data, including locations, organizations, or specific entities pertinent to airport operations. NER aids in extracting crucial information and pinpointing potential risk factors or entities linked with safety incidents or operational disruptions.

Text classification algorithms, like Support Vector Machines (SVM) [13] or Naive Bayes classifiers [14], categorize textual data into predefined classes or categories based on their content or context. This categorization assists in assigning risk levels or consequence categories to reported incidents, enhancing the prioritization and allocation of resources for risk mitigation.

Semantic analysis delves into the underlying meaning and context of textual data, uncovering implicit relationships and insights. By deciphering the semantic structure of textual content, semantic analysis techniques unearth patterns or associations indicative of potential risks or safety concerns.

In summary, the integration of text analysis techniques empowers stakeholders in airport operation assurance to extract actionable insights from unstructured textual data. By employing sentiment analysis, topic modeling, named entity recognition, text classification, and semantic analysis, stakeholders can bolster their capacity to identify and address operational risks effectively, thus enhancing the safety and resilience of airport operations.

D. Introduction to the DBSCAN Algorithm and Its Application in Clustering Textual Data

The Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm is a fundamental clustering technique widely used in data mining and machine learning. Introduced by Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu in 1996, DBSCAN has gained prominence due to its ability to efficiently cluster spatial data points based on their density distribution. Unlike traditional clustering algorithms, such as k-means, DBSCAN does not require a predefined number of clusters and is capable of identifying clusters of arbitrary shapes and sizes. This section provides an overview of the DBSCAN algorithm and its application in clustering textual data.

The DBSCAN algorithm operates by partitioning the input data space into clusters, where each cluster consists of data points that are densely packed together. The key concept underlying DBSCAN is the notion of density reachability, which defines clusters as regions of high density separated by areas of low density. DBSCAN classifies data points into three categories: core points, border points, and noise points, as shown in Fig. 3.

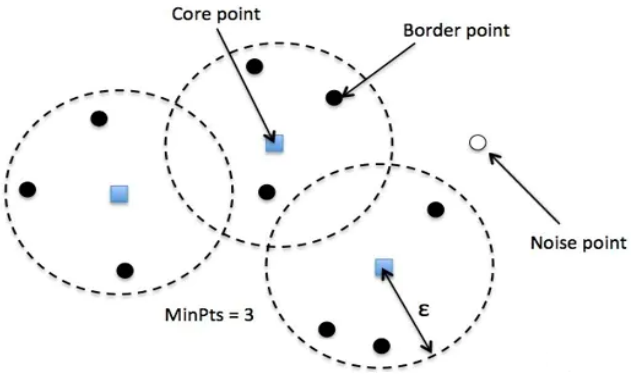


Fig. 3. Three categories of points of the DBSCAN algorithm.

Core Points: A core point is a data point with a specified minimum number of neighboring points (MinPts) within a specified radius (ε).

Border Points: Border points are not core points themselves but belong to the neighborhood of a core point. They have fewer neighboring points than required to be considered core points.

Noise Points: Noise points do not belong to any cluster and are not core points or border points. They lie in regions of low density and do not satisfy the density criteria to be considered core points.

The DBSCAN algorithm is executed in the following steps:

Step 1: Input Parameters

ε (Epsilon): This parameter defines the maximum distance between two points for them to be considered neighbors.

MinPts: The minimum number of points needed to form a dense region (cluster).

Step 2: Neighborhood Query

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For each data point in the dataset, the algorithm calculates its ε-neighborhood, encompassing all points lying within a distance ε from the current point.

Step 3: Core Point Identification

Core points are identified by assessing the number of points in their ε-neighborhood. If this count exceeds or equals MinPts, the point is deemed a core point.

Step 4: Cluster Expansion

The algorithm proceeds to recursively expand clusters by incorporating all directly reachable points (i.e., those within the ε-neighborhood) into the cluster. If a neighboring point is itself a core point, its ε-neighborhood is subsequently explored.

Step 5: Border Points Assignment

Border points are assigned to the clusters of their corresponding core points. These points, while not core points themselves, are reachable from core points within the ε-neighborhood.

Step 6: Noise Points Identification

Noise points, which fails to meet the density criteria and are not part of any cluster, are identified as outliers. These points are typically characterized by their inability to belong to a cluster due to their isolated nature.

The DBSCAN algorithm flow is shown in Fig. 4.

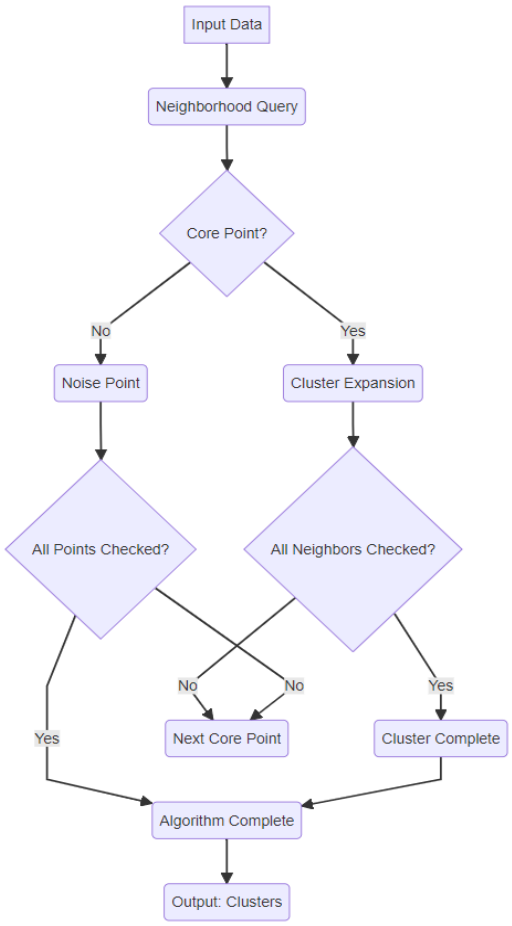


Fig. 4. The schematic flow of the DBSCAN algorithm .

The time complexity of DBSCAN is contingent upon various dataset attributes, including size, dimensionality, and data point distribution. In the worst-case scenario, DBSCAN exhibits a time complexity of O(n^2), where n denotes the number of data points.

However, in practice, DBSCAN often showcases greater efficiency than its theoretical worst-case complexity suggests, especially when confronted with datasets characterized by high density and low dimensionality.

As mentioned above, the DBSCAN algorithm embodies a robust and efficient approach to clustering data points based on their density distribution. By delineating core points, expanding clusters, and identifying noise points, DBSCAN adeptly partitions datasets into clusters of varying shapes while adeptly handling noise and outliers.

III. Methodology

A. Data Preprocessing of Voluntary Reports

Text preprocessing plays a pivotal role in preparing textual data for subsequent analysis tasks. In this section, the scholarly methodology adopted for text preprocessing was delineated, leveraging Python's Natural Language Toolkit (NLTK) library, and conducting feature extraction through the utilization of scikit-learn's TfidfVectorizer.

1) Text Preprocessing with NLTK

Stopword Removal: NLTK provides a comprehensive corpus of stopwords in various languages. These stopwords, common words such as "the," "is," and "and," do not contribute significantly to the semantics of the text and are therefore typically removed to improve analysis accuracy.

Tokenization: Tokenization involves segmenting text into individual words or tokens. NLTK offers robust tokenization utilities, including word_tokenize, which breaks text into tokens based on whitespace and punctuation.

Lowercasing: Converting all text to lowercase ensures uniformity in textual representation and simplifies subsequent analysis tasks. NLTK's string manipulation functionalities facilitate efficient lowercase conversion.

2) Feature Extraction with TfidfVectorizer

Term Frequency-Inverse Document Frequency (TF-IDF): TfidfVectorizer, a component of scikit-learn's feature extraction module, transforms text documents into numerical feature vectors based on the TF-IDF weighting scheme. This scheme accounts for both the frequency of a term within a document (TF) and its inverse frequency across all documents (IDF).

Stopword Removal and Tokenization Integration: TfidfVectorizer seamlessly integrates with NLTK's stopword removal and tokenization functionalities. By specifying NLTK's stopwords as parameters and utilizing NLTK's tokenization methods, TfidfVectorizer ensures the extraction of meaningful features while circumventing the inclusion of stopwords and irrelevant tokens.

Efficient Feature Vectorization: TfidfVectorizer efficiently converts a collection of text documents into a sparse matrix representation, where each row corresponds to a document and each column represents a unique term in the corpus. This matrix encapsulates the TF-IDF weighted features, which serve as input for subsequent analysis tasks,

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such as clustering or classification.

Through the judicious utilization of NLTK for text preprocessing and scikit-learn's TfidfVectorizer for feature extraction, researchers can effectively transform raw textual data into structured, numerical representations conducive to a wide array of text mining and natural language processing tasks. These methodologies underscore the foundational steps in harnessing textual data for insightful analysis and interpretation. The sample code for the above procedure is shown in Fig. 5.

```
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer

# Download NLTK stop words and tokenizers
nltk.download('stopwords')
nltk.download('punkt')

# Perform text preprocessing and feature extraction on the report
def preprocess_and_extract_features(reports):
    # Initialize the stop word collection
    stop_words = set(stopwords.words('english'))

    # Initialize the TF-IDF vectorizer
    tfidf_vectorizer = TfidfVectorizer(stop_words=stop_words)

    # Tokenization and removal of stop words
    tokenized_reports = [' '.join(word_tokenize(report.lower())) for report in reports]

    # Use the TF-IDF vectorizer to extract features
    features = tfidf_vectorizer.fit_transform(tokenized_reports)

    return features

# Text preprocessing and feature extraction were performed on voluntary reporting datasets
preprocessed_features = preprocess_and_extract_features(voluntary_reports)
print(preprocessed_features.shape)  # Prints the shape of the feature matrix
```

Fig. 5. The code of data preprocessing module.

B. Implementation of DBSCAN clustering algorithm

This section outlines the implementation of DBSCAN clustering algorithm for clustering preprocessed textual data.

1) DBSCAN Clustering

Parameter Selection: Before applying DBSCAN, it's crucial to choose appropriate values for the parameters ε (epsilon) and MinPts (minimum number of points). These parameters influence the clustering results and need to be adjusted based on the characteristics of the data.

Initialization: Initialize DBSCAN with the chosen ε and MinPts parameters.

Fit Data: Apply DBSCAN to the preprocessed feature matrix using the fit method.

2) Incorporating Cluster Labels

Assigning Cluster Labels: Once DBSCAN is fitted to the data, cluster labels are assigned to each data point based on the clustering results.

Adding Cluster Labels: Add the cluster labels to the original DataFrame to associate each report with its corresponding cluster.

3) Displaying Clustering Results

Printing Clustered Reports: Iterate through unique cluster labels and print reports belonging to each cluster, facilitating the examination of the clustering results.

4) Visualization (Optional)

Scatter Plot: If there are sufficient features available for visualization, a scatter plot can be generated to visualize the clustering results. However, given the high-dimensional nature of textual data, dimensionality reduction techniques such as Principal Component Analysis (PCA) are often employed to visualize clustering results in two or three dimensions. The sample code of the DBSCAN clustering algorithm is illustrated in Fig. 6.

```
# Implementation of DBSCAN clustering algorithm

# Parameter Selection
eps = 0.5  # ε parameter (need adjustment based on data)
min_samples = 5  # MinPts parameter (need adjustment based on data)

# Initialization
dbscan = DBSCAN(eps=eps, min_samples=min_samples).fit(preprocessed_features)

# Incorporating Cluster Labels
df_reports['Cluster'] = dbscan.labels_

# Displaying Clustering Results
for cluster_label in np.unique(dbscan.labels_):
    print("Cluster", cluster_label, ":", df_reports[df_reports['Cluster'] == cluster_label])

# Visualization (Optional)
reduced_data = PCA(n_components=2).fit_transform(preprocessed_features.toarray())
plt.scatter(reduced_data[:, 0], reduced_data[:, 1], c=dbscan.labels_)
plt.title('DBSCAN Clustering')
plt.xlabel('PCA Feature 1')
plt.ylabel('PCA Feature 2')
plt.show()
```

Fig. 6. The implementation code of the DBSCAN clustering algorithm.

This implementation showcases the application of DBSCAN clustering algorithm to preprocessed textual data, enabling the identification of clusters and exploration of the underlying structure within the dataset. Adjusting the ε and MinPts parameters according to the data characteristics is essential to achieve meaningful clustering results.

IV. Experiments

A. Dataset

In this paper, a dataset was constructed consisting of 100 voluntary reports, each of which detailed key information such as report number, content description, and report date. Considering the sensitivity of the DBSCAN algorithm to high-dimensional data, the content of the report is processed through text vectorization, employing the TF-IDF (Term Frequency-Inverse Document Frequency) method to transform textual data into numerical feature vectors. Furthermore, in order to mitigate the influence of the proportion between different features and the effect of data display, the dataset will be standardized into two different sets of 2D feature vectors according to the likelihood and consequence.

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B. Evaluation of risk likelihood

This paper categorizes the likelihood of risk into five distinct levels: Almost Certain, Likely, Possible, Unlikely, and Rare. A standard sample is predefined for each category, with an average of 20 samples per category, thereby allowing the setting of MinPts to 5 (5% of the samples). The value of ε, however, necessitates empirical testing. The clustering outcomes are presented when ε is set within the range of 0.3 to 0.8, as depicted in Fig. 7.

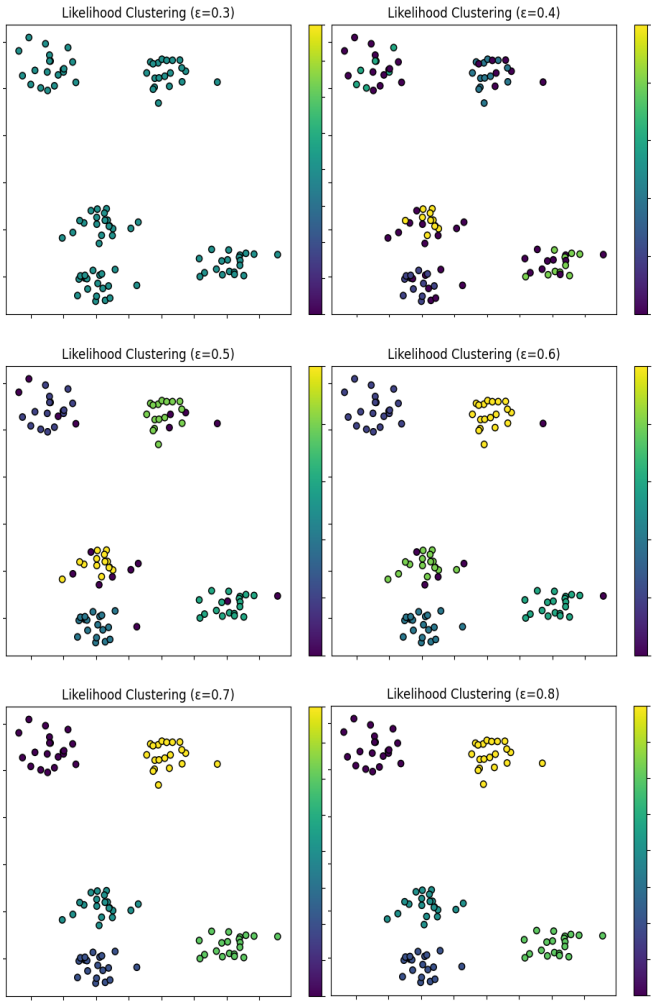


Fig. 7. Likelihood Clustering outcomes.

It is observable that when ε is set to 0.3, the entire dataset converges into a single cluster, which is evidently suboptimal. As the value of ε increases, the clustering outcomes progressively improve. At ε=0.6, discrete points persist. Beyond ε≥0.7, the clustering results stabilize. Consequently, the optimal range for ε is determined to be (0.6, 0.7].

C. Evaluation of risk consequence

Similar to the categorization of risk likelihood, this paper delineates the consequences of risk into five distinct levels: Insignificant, Minor, Moderate, Major, and Severe. A standard sample has been predefined for each category, with MinPts set to 5. Correspondingly, the value of ε is subject to empirical validation. A comparative analysis of clustering outcomes at various ε values is shown in Fig. 8.

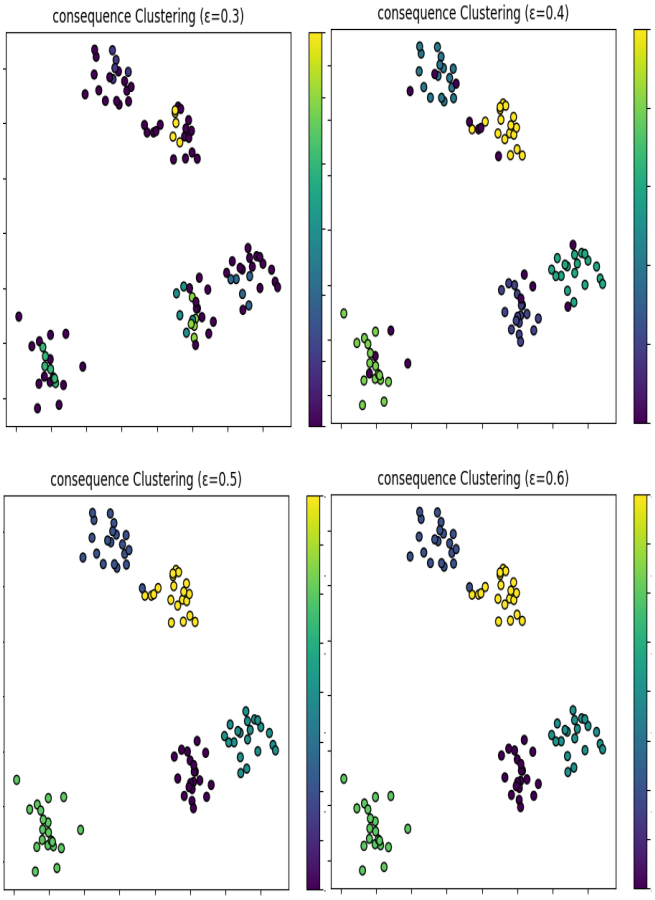


Fig. 8. Consequence Clustering outcomes.

It is discernible that, in contrast to the risk likelihood, the dataset can already form two distinct clustering centers when ε is set at 0.3. As the value of ε increases, the clustering outcomes are observed to progressively refine. At ε=0.4, discrete points continue to persist. Once ε exceeds the threshold of 0.5, the clustering results achieve stability. Consequently, the optimal range for ε is ascertained to be (0.4, 0.5].

D. Determination of risk levels

This paper delineates four categories of risk levels: Low, Medium, High, and Extreme, represented respectively by green, yellow, brown, and red. The risk levels are ascertained through a risk matrix as depicted in Fig. 9, which integrates various combinations of likelihood and consequence.

Likelihood	Consequences				
	Insignificant	Minor	Moderate	Major	Severe
Almost Certain	M	H	H	E	E
Likely	M	M	H	H	E
Possible	L	M	M	H	E
Unlikely	L	M	M	M	H
Rare	L	L	M	M	H

Fig. 9. Risk matrix.

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Initially, clustering is performed on both likelihood and consequence, and the outcomes are aligned with the predefined anchor values. This process yields the likelihood and consequence values for each sample data point. Subsequently, by applying the aforementioned risk matrix, the risk level of the sample can be determined.

V. Conclusions

The principal contribution of this study lies in the successful application of the DBSCAN algorithm to the assessment of operational risk in airport environments. By leveraging text analysis and clustering techniques, a novel approach for airport risk management has been provided. Initially, a preprocessing framework for voluntary reporting textual data was developed, effectively extracting salient features from raw text and laying the groundwork for subsequent cluster analysis. Subsequently, a parameter optimization strategy was proposed, integrating domain knowledge and data characteristics, thereby enhancing the DBSCAN algorithm's adaptability to the assessment of operational risks in airport environments. Finally, a risk matrix was designed, capable of calculating the risk categories of samples from clustering results, thereby offering data support for risk management and decision-making. Through experimentation, the feasibility of the methods designed in this study has been validated. Future work could involve further verification of the clustering model's effectiveness through case studies and adjustment and optimization of model parameters based on practical application scenarios.

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